Insect/Pest Control Methods

➢ Insect:

Any organism which has three pairs of legs(6 legs) and body divided into three parts i-e "Head, Thorax & Abdomen".

➢ Pest:

Any organism that damages crops, injures or irritates livestock or man, or reduces the fertility of land (Broader term).

➢ Conrol:

The killing, reduction or elimination of insects which are pests, using insecticide, etc to avoid damage.

➢ Methods:

- The methods should be most effective & less harmful to environment(Humans & animals).
- Control methods includes everything that makes life hard for insects.
- Control methods also tend to kill insects to prevent their increase in particular area.

➢ Insects are kept under control in two ways:

I. NATURAL CONTROL:

- The measures that distroy or check insects without the influences of the man.
- This includes: Climatic, topographic & biotic factors.

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a) Climatic factors:

- Include rainfall, cold, heat and wind.
- These climatic factors effect the insect population to great extent.
- Rainfall and hail storm kill insect in larval stage.
- Extreme heat and cold temperature affect a number of insects spp.
- Wind movement is also of great importance i-e many spp of smaller insects which normally fly for considerable distance are beaten by the wind.

b) Topographic factors:

- Include rivers, lakes, mountains, types of soil & other barriers.
- Rivers and lakes affect those insects which do not possess the power of flight.
- Mountains offer varying conditions of climatic through which many insects cannot pass.
- The character of soil of any region affects a marked influence on the insects.
  - For example; Certain spp of tiger beetle larvae live in sandy soil and are unable tp exist in the clay soil.

c) Biotic factors:

- Include predatory insects, birds, reptiles, mamales, parasitic insects, fungi, bacteria & other microorganasms.
- A large number of insects parasite on other insects.
- Some are predators of insects.
- Almost all birds feed on adult insects as well as other worms.
- Insects are also attacked by fungi, bacterial & viral diseases.

II) APPLIED CONTROL:

•To distroy insects by the uses of man made methods.

• This includes: Silvicultural, biological, mechanical, physical & chemical control methods.

•Another metod is "Integrated pest management(IPM)" is new concept of insect control.

•IPM minimizes harm to the environment while keeping problem insects in check.

➢Silviculture methods include the practices which distroy insects i-e mixed croping & tending etc.

➢Biological method is the control of insects by other livings i-e predation.

➢Mechanical method is the control of insects through mechanical means i-e collection of insects by hand.

➢Physical method is the control of insects by the increase or decrease in temperature, humidity & wind.

➢Chemical method is the control of insects through some chemicals i-e insecticides, herbicides, pesticides etc.

❑ Other methods of insect control include:

a. Raising healthy nurseries:

- Diseased free seeds should be collected from healthy trees & raised in nurseries.
- Excellent preparation of seeds beds and proper irrigation & fertilizing in the nurseries should be ensured from healthy nurseries.

b. Mixed forest:

- Mixed forests prevent the spread of specific insects.

➢ For example; Shisham defoliator can be controled by planting it with mulberry.

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c) Removing over aged trees from forests:

- Over aged trees usually provide breeding place for bark and stem borers.
- A number of defoliators usually attack the barks of old and weakend trees.

➢ For example; Poplar borer breed well in the aged trees.

□ Removal of the old aged and weakend trees & their stumps minimize the outbreak of the borers.

d) Debarking of felled trees:

- Logs of pines trees should be debarked just after felling to avoid the attack of bark beetles.

e) Phase wise felling:

- A large clear felled area is more attractive to forest pest than the small area.
- A phase wise felling should be under taken in small patches.

➢ Artificial or natural regeneration should be ensured in felled area.

❑ This will also control many pests

f) Control burning:

- Large scale burning in the natural forest particularly young plantation should be avoided.
- Burning encourages the bark beetle and also removes the natural enemies of the pests.

➢ Control burning should be practiced carefully particularly in young plantations.